

Faculté des sciences et technologies

Le rapport du quatrième laboratoire du réseaux I

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Discipline : Réseaux I

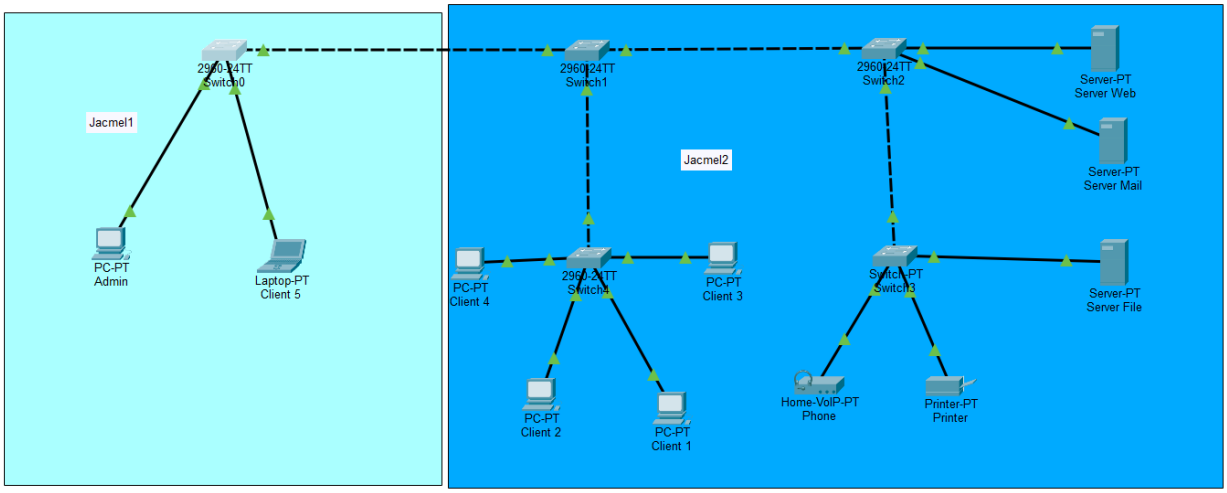
Date : 12 Novembre 2026

L’objectif de ce TD est de :

1. Découvrir la configuration de base d’un switch et d'un routeur.
2. Attribuer des adresses IPv4 aux périphériques.
3. Tester la connectivité entre les hôtes.
4. Utiliser le Mode Simulation pour analyser le trafic réseau

Travaux Dirigés

1.Reproduction d'une topologie contient des PC, des Switchs, des servers, printer, Laptop puis un phone.



La configuration des Switchs

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#interface vlan 1
SW2(config-if)#ip address 192.168.1.254 255.255.255.0
SW2(config-if)#no shutdown

SW2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SW2(config-if)#exit
SW2(config)#ip default-gateway 192.168.1.255
SW2(config)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW3
SW3(config)#interface vlan 1
SW3(config-if)#ip address 192.168.1.255 255.255.255.0
Bad mask /24 for address 192.168.1.255
SW3(config-if)#ip address 192.168.1.25 255.255.255.0
SW3(config-if)#no shutdown

SW3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SW3(config-if)#exit
SW3(config)#ip default-gateway 192.168.1.256
                               ^
% Invalid input detected at '^' marker.

SW3(config)#ip default-gateway 192.168.1.256
                               ^
% Invalid input detected at '^' marker.

SW3(config)#ip default-gateway 192.168.1.25
SW3(config)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW4
SW4(config)#interface vlan 1
SW4(config-if)#ip address 192.168.1.26 255.255.255.0
SW4(config-if)#no shutdown

SW4(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SW4(config-if)#exit
SW4(config)#ip default-gateway 192.168.1.26
SW4(config)#end
SW4#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW5
SW5(config)#interface vlan 1
SW5(config-if)#ip address 192.168.1.27 255.255.255.0
SW5(config-if)#no shutdown

SW5(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

SW5(config-if)#exit
SW5(config)#ip default-gateway 192.168.1.27
SW5(config)#end
SW5#
%SYS-5-CONFIG_I: Configured from console by console
```

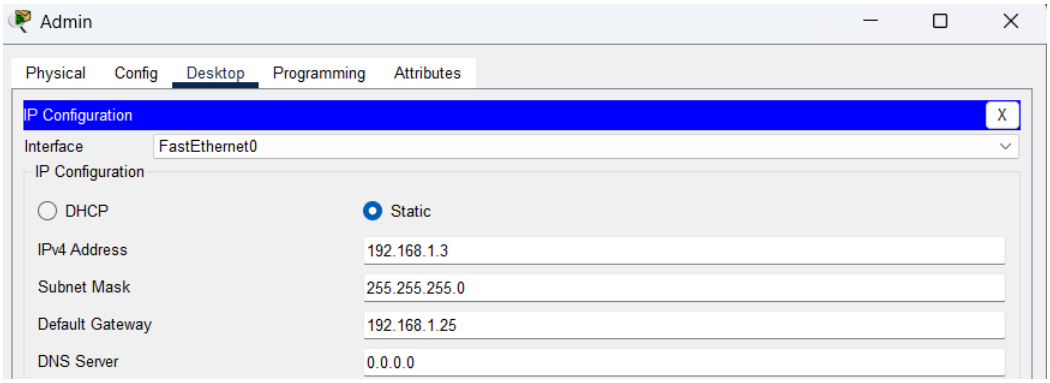
```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW5
SW5(config)#interface vlan 1
SW5(config-if)#ip address 192.168.1.27 255.255.255.0
SW5(config-if)#no shutdown

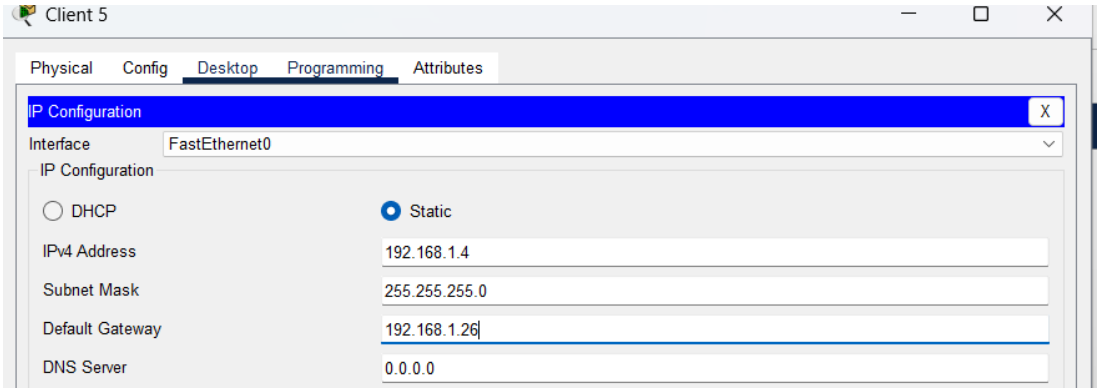
SW5(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

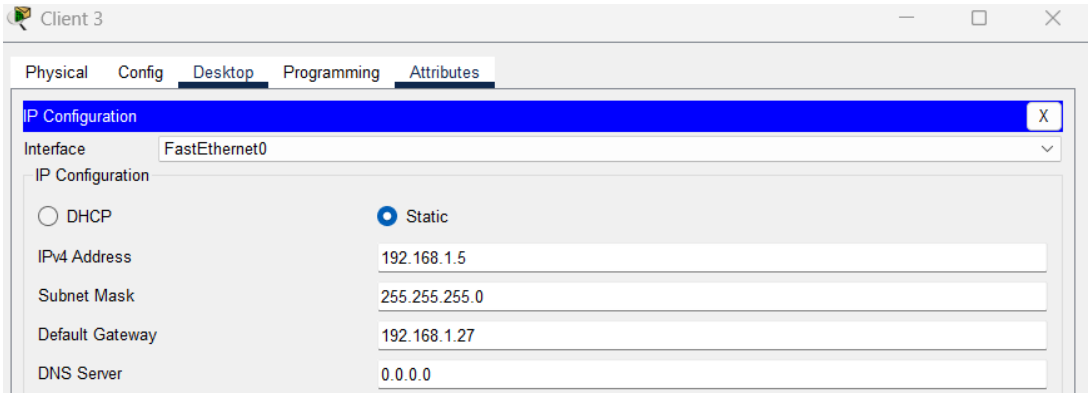
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

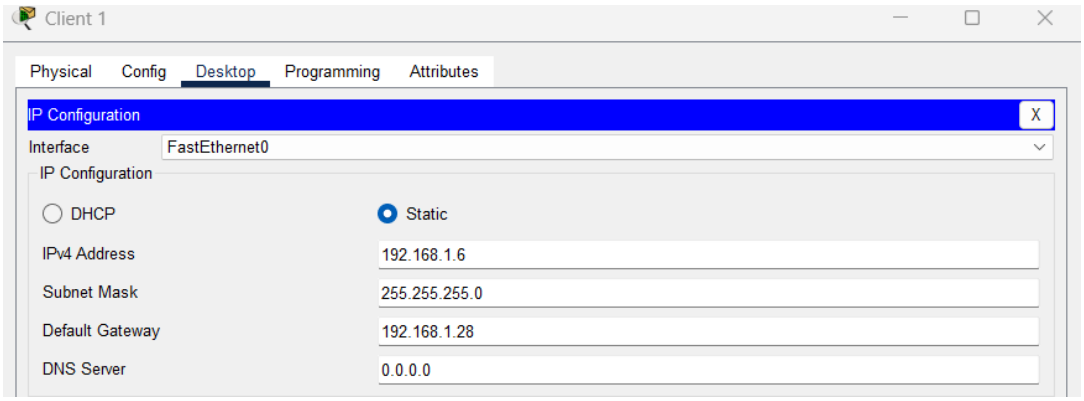
SW5(config-if)#exit
SW5(config)#ip default-gateway 192.168.1.27
SW5(config)#end
SW5#
%SYS-5-CONFIG_I: Configured from console by console
```

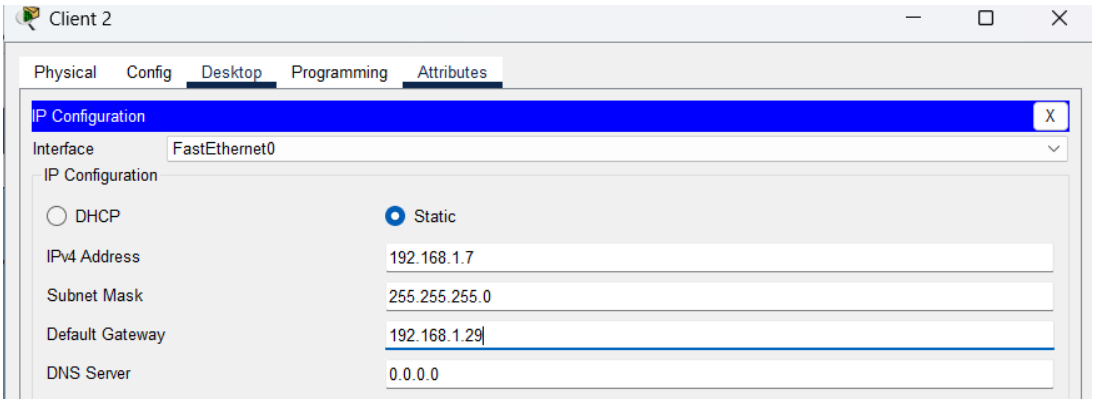
La configuration des PC

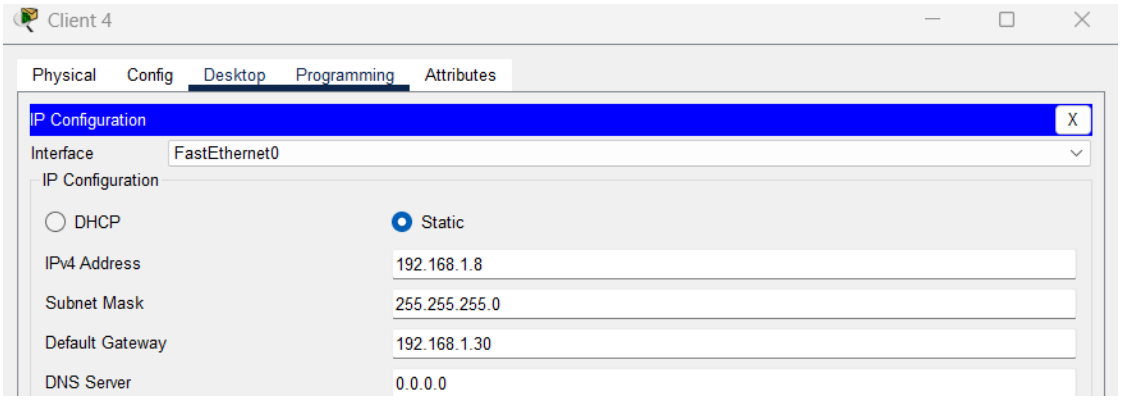



Client 5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.4
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.26
DNS Server
0.0.0.0

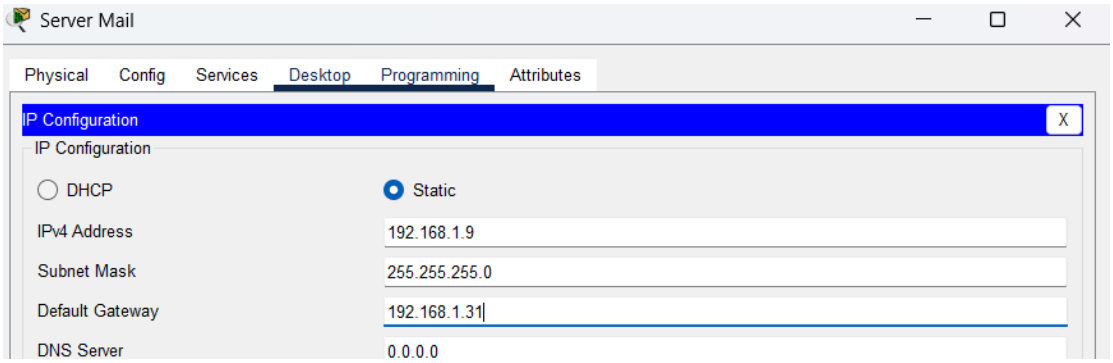

Client 3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.5
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.27
DNS Server
0.0.0.0

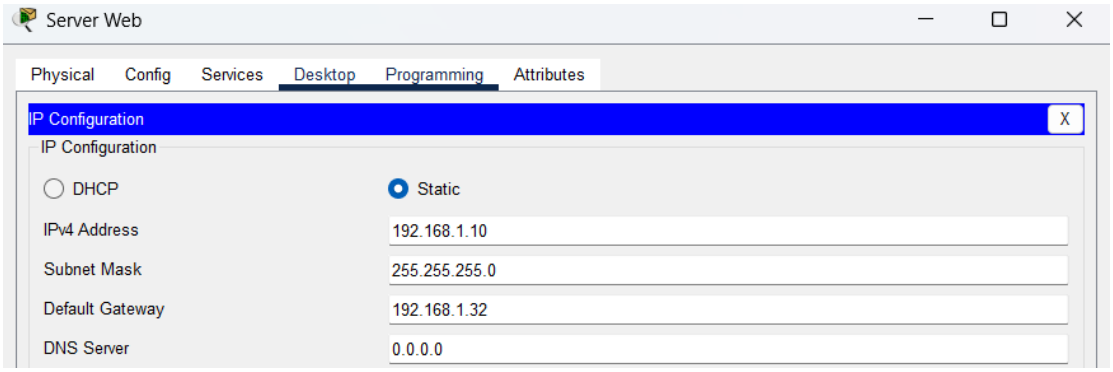

Client 1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.6
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.28
DNS Server
0.0.0.0

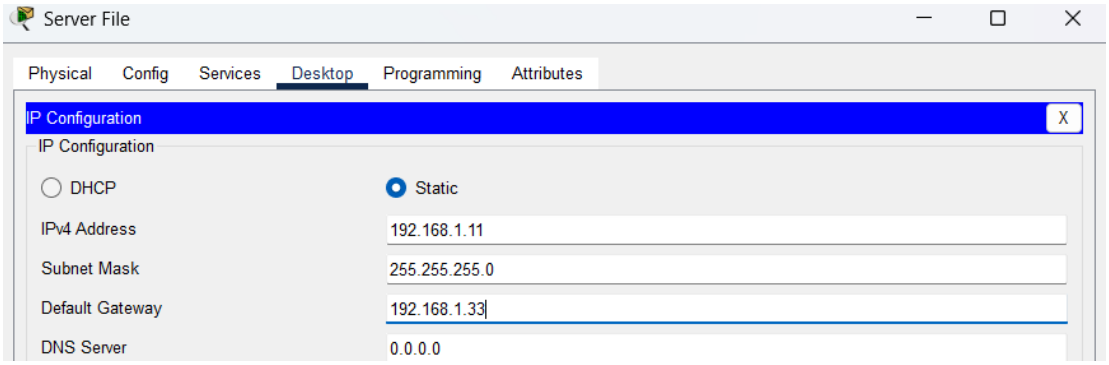

Client 2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.7
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.29
DNS Server
0.0.0.0



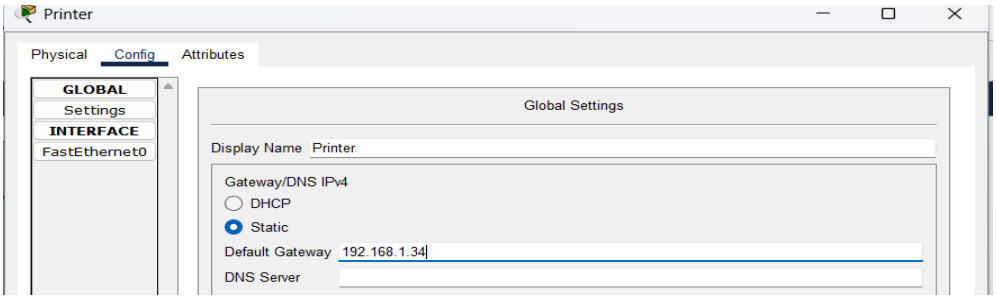
Configuration des servers



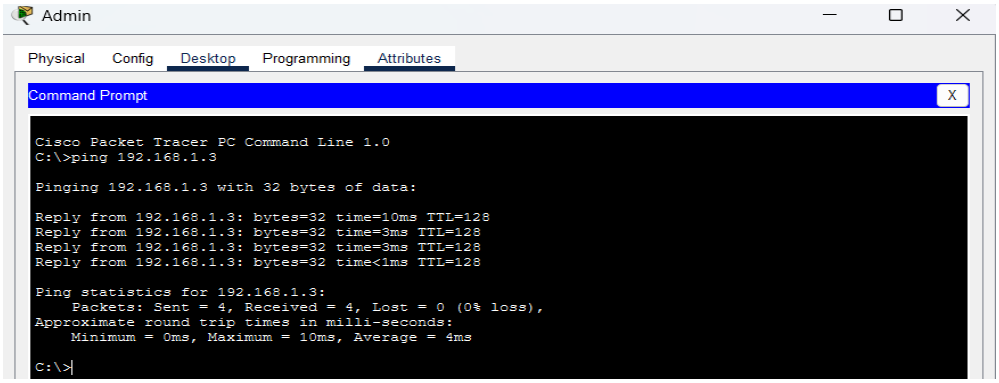




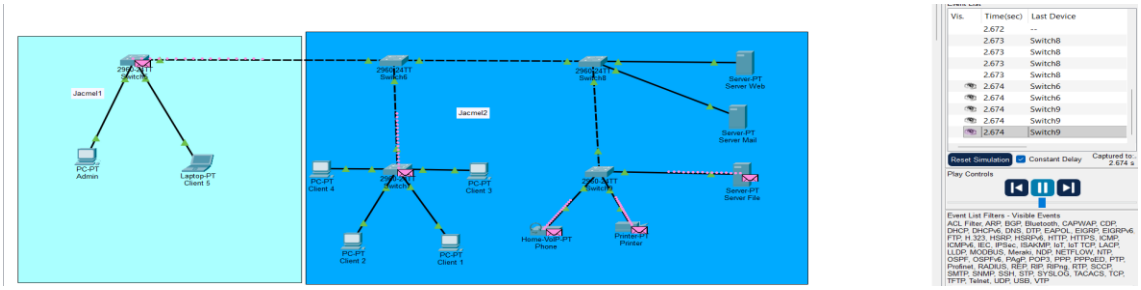
Configuration du printer



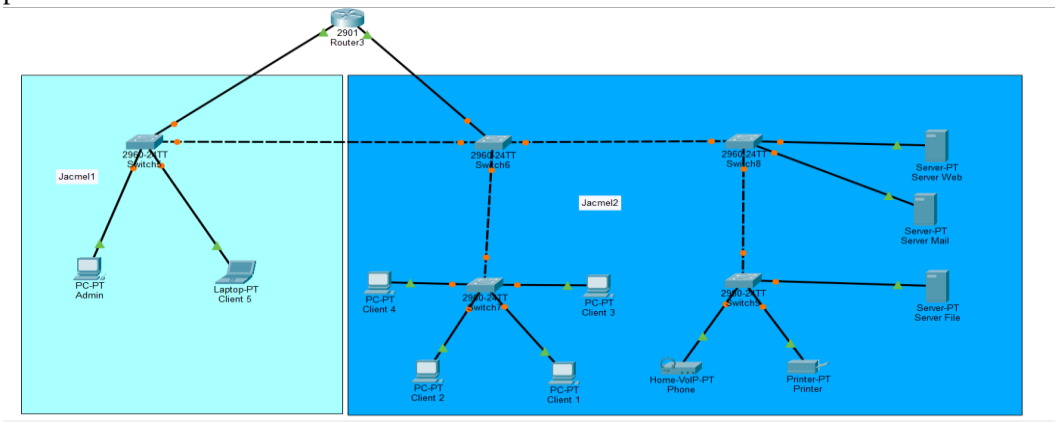
Test de connectivite avec la command ping plus address IPv4 d'un PC



La simulation



2- La reproduction d'une topologie contient des PC, des Switchs, des servers, printer, Laptop, un phone puis un Routeur modèle 2901.



La configuration du routeur

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface FastEthernet0/0
%Invalid interface type and number
R1(config)#interface FastEthernet0/0
%Invalid interface type and number
R1(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip address 192.168.1.250 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip address 192.168.1.354 255.255.255.0
                         ^
% Invalid input detected at '^' marker.

R1(config-if)#ip address 192.168.1.251 255.255.255.0
% 192.168.1.0 overlaps with GigabitEthernet0/0
R1(config-if)#ip address 192.168.1.250 255.255.255.0
% 192.168.1.0 overlaps with GigabitEthernet0/0
R1(config-if)#ip address 192.168.2.250 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
```

```
R1(config-if)#ip address 192.168.1.251 255.255.255.0
% 192.168.1.0 overlaps with GigabitEthernet0/0
R1(config-if)#ip address 192.168.1.250 255.255.255.0
% 192.168.1.0 overlaps with GigabitEthernet0/0
R1(config-if)#ip address 192.168.2.250 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.250   YES manual up                    up
GigabitEthernet0/1     192.168.2.250   YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Le test de connectivite avec la commande ping et l'address IPv4 d'un PC

```
C:\>ping 192.168.1.3

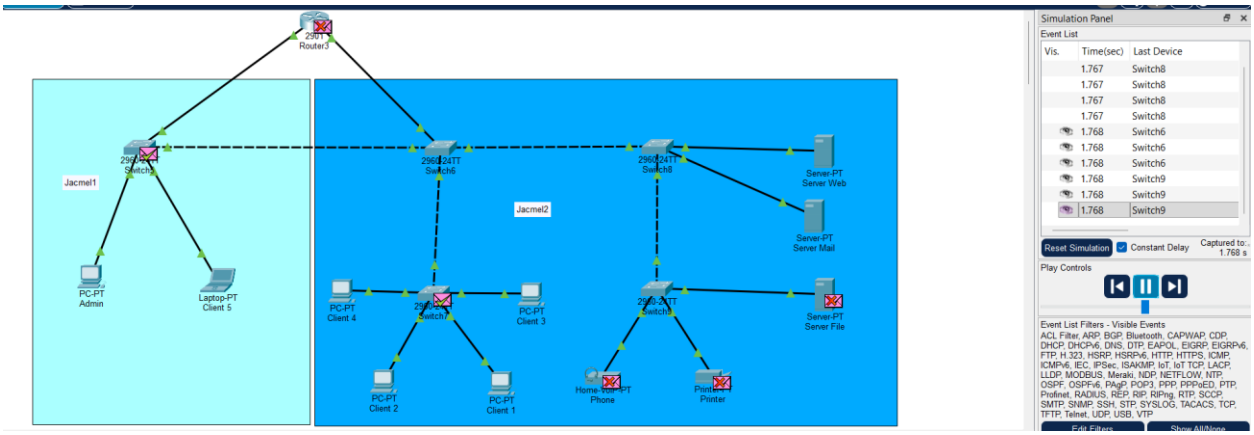
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=3ms TTL=128
Reply from 192.168.1.3: bytes=32 time=3ms TTL=128
Reply from 192.168.1.3: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>
```

La simulation de cette topologie



Conclusion

Ce TD m'a appris beaucoup de choses sur la connexion des appareils entre eux, ainsi que sur la configuration du routeur et des Switchs. Malgré je pas encore maitrise le mode simulation mais j'ai réussi à le faire quand même. Ce TD m'a permis de comprendre les différents interfaces d'un routeur ne peuvent pas appartenir au même réseau : lorsqu'une interface utilise l'address 192.168.1.250 avec le masque 255.255.255.0 l'autre ne peut pas prendre cette même plage d'address, mais doit utiliser une autre tel que 192.168.2.250 255.255.255.0, car la première la plage est déjà attribuée.